Supplementary Material for CLADE: Category-Level Answering from Deficient Evidence in Retrieval-Augmented Language Models

Anonymous submission

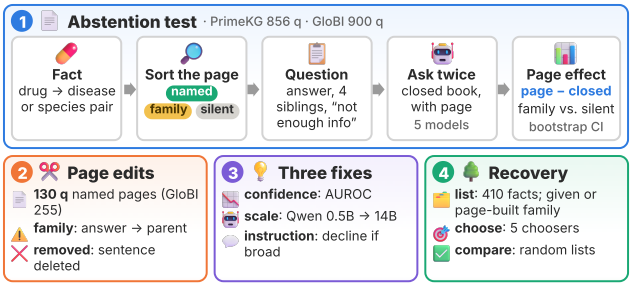


Figure 1: Our four experiments. (1) We sort each page as named, family-only or silent, and ask each question closed book and with the page. (2) We edit named pages: the answer becomes its family name, or we delete the sentence. (3) We test three fixes. (4) We try to recover the exact answer from the ontology.

Main paper	Supplement
Data, page types, Tables 1 and 3	Appendix B
Odds ratios, CIs, p-values, Eq. 1	Appendix C
Questions, prompts, Table 4, Figure 2	Appendix D
Page edits, Figure 3	Appendix E
Confidence, scale, instruction, Figure 4	Appendix F
GloBI, Table 5	Appendix G
Recovery, Table 6, Figure 5	Appendix H
Worked examples	Appendix I
Software and compute	Appendix J
Choosing the family, free text	Appendix K

Table 1: Where each part of the main paper is explained in this supplement.

A Overview and Reproducibility

This supplement gives the details behind each part of the main paper. It follows the same order as the paper: data and page types, statistics, the abstention test, page edits, the three fixes, the second domain, and ontology recovery. The last appendix covers two follow-up tests: a model that chooses the family on the page, and free-text answers. Table 1 shows where each result of the main paper is explained in more depth. Figure 1 summarizes the four experiments in one picture.

Code and data. We release all code and derived data at https://anonymous.4open.science/r/CLADE. The release holds the question sets, the windowed pages, the candidate lists, the per-question model outputs, and every script that builds them. We do not redistribute Wikipedia or PubMed text; the scripts download it again from the public APIs. PrimeKG, MONDO, GloBI, Wikipedia and PubMed are all public.

Rerunning the numbers. Every table and figure comes from the shipped per-question outputs, so a reader can check them without a GPU or network access:

```
bash tools/check_results.sh
# reruns the analysis and compares it
# with results/; prints "identical: 22"
```

The full pipeline, from raw downloads to the final tables, runs with one command on a SLURM cluster:

```
bash run_all.sh              # PrimeKG
bash globi.sh all             # GloBI
```

Both scripts resume where they stop, and a smoke test on small synthetic data checks the whole pipeline in a few minutes (`-smoke`).

B Data and Page Types

This appendix explains how we build the two graphs, how we read the text, and how we decide whether a page names the answer, names only its family, or says nothing.

B.1 Graphs and Hierarchies

PrimeKG and MONDO. PrimeKG has 8 100 498 rows. We keep three drug–disease relations: 9388 indications, 30 675 contraindications and 2568 off-label uses. Each disease has a MONDO identifier. We rebuild MONDO as a table of child–parent links: 17 801 links over 11 788 diseases. A parent with two or more children is a *family*; MONDO has

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3182 of them. We drop parents with more than 400 descendants, such as *disease* or *disorder*. Naming them on a page tells the reader almost nothing, and they would turn almost every page into a family-only page.

GloBI and its taxonomy. We use the archived GloBI dump (Zenodo record 4460654), which holds 7 822 670 interaction records. We keep four relations: *eats*, *preys on*, *parasite of* and *pollinates*. Together they hold 3 210 597 records. We keep a record only when both sides are species, written as a two-word Latin name ("Genus epithet"). We drop names with "sp.", hybrids and numbers, because a question about a genus has no single answer. This leaves 457 386 species-to-species edges:

Relation	Edges	Source species
eats	235 960	25 637
parasite of	117 894	31 432
preys on	87 678	5138
pollinates	15 854	2818

GloBI records give a taxonomic path for each taxon (species, genus, family, order, class). We build the hierarchy from these paths. Different records can give a taxon different parents, so we keep the parent seen most often. This makes the table a tree: climbing it never moves sideways. The result has 77 852 child–parent links, and 8370 genera have two or more species. As in MONDO, the median family has three members.

B.2 Text Sources

Wikipedia. For each drug, and for each consumer species in GloBI, we fetch the English Wikipedia article through the MediaWiki API. We turn the raw markup into plain prose. We remove comments, reference tags, templates (including infoboxes and citation blocks, with nested braces), tables, and file and category links. We replace each internal link with its visible text and remove bold and italic marks, HTML tags and character entities. We keep section headings as plain lines, so the text reads like the page a person sees. For GloBI we skip stubs shorter than 3000 characters.

PubMed. For each drug we retrieve PubMed abstracts through NCBI E-utilities. We split the query budget equally between a treatment query and an adverse-effect or contraindication query, so both relations get the same chance to appear. The drug's full text is its Wikipedia article followed by its abstracts.

B.3 How a Page Names a Fact

Three name sets. Each answer has three nested sets of names:

- the *exact name*: its primary MONDO label, or its Latin name in GloBI;
- *alternative names*: titles of Wikipedia pages that redirect to the answer's article, which bring in synonyms, lay terms and abbreviations; in GloBI we also add the abbreviated form (*A. plantaginifolia*);
- *family names*: the labels of the answer's ancestors up to two levels above it (the MONDO parent and grandparent, or the genus and family).

	True disease		Unrelated diseases	
Relation	named	+ family	named	+ family
Indication	19.3	34.7	1.4	2.7
Contraindication	6.7	9.3	0.9	1.8
Off-label use	11.8	23.7	0.9	3.3

Table 2: Share of facts (%) whose Wikipedia text matches the true disease or one of three unrelated diseases from the same relation (600 facts per relation; 1800 unrelated draws each). "+ family" also counts family names.

Page type. Let T be the text, $N(a)$ the exact and alternative names of answer a, and $A_2(a)$ its family names. We assign each fact to the finest level the text matches:

$$\text{type}(T,a) = \begin{cases} \text{named} & \text{if } T \text{ matches } N(a), \\ \text{family-only} & \text{else if } T \text{ matches } A_2(a), \\ \text{silent} & \text{otherwise.} \end{cases} \quad (1)$$

Matching rule. A name matches when it appears on word boundaries, ignoring case, with an optional trailing *s* for simple plurals. A multi-word name matches across any run of white space. We ignore names shorter than four characters, because short acronyms match by accident.

Chance control. A family name is broader than a disease name, so it can match unrelated text more easily. To measure this, we pair each fact with three diseases from the same relation that have no link to the drug, and we run the same matcher on the same text. Table 2 compares the true disease with these unrelated ones on 1,800 PrimeKG facts in Wikipedia. The unrelated diseases match 0.9–1.4% of the time at the named level and 1.8–3.3% once family names are added. The true disease matches 6.7–19.3% and 9.3–34.7%. Matches therefore follow the fact, not the size of the name set.

B.4 Text Coverage and Closed-Book Knowledge

The four-box table. Table 2 in the main paper uses 1,800 PrimeKG facts (600 per relation). A fact counts as *text names it* when the drug's Wikipedia article contains the exact or an alternative name. A family-only page counts as *text does not name it*. To test closed-book knowledge, Qwen2.5-0.5B scores the true disease against four random diseases from the same relation. It reads a short stem, such as "*X is a drug used to treat the disease called*", and we take the mean log-probability of each disease name after it. The highest score is the model's pick.

Knowledge above chance. On facts the text does not name, the model picks the true disease 25.1% of the time. With five options, chance is 20%. We correct for guessing with

$$k = \frac{\text{acc} - c}{1 - c} = \frac{0.251 - 0.200}{1 - 0.200} \approx 6.4\%, \quad (2)$$

Relation	Named	Family only	Silent
Indication	84.2 (101)	44.7 (94)	44.0 (405)
Contraind.	32.5 (40)	18.8 (16)	23.4 (544)
Off-label	57.1 (63)	37.5 (72)	31.6 (465)
All	65.7 (204)	39.6 (182)	32.0 (1,414)

Table 3: Closed-book accuracy of Qwen2.5-7B by what the drug's Wikipedia page says about the answer. Chance is 20%.

where c is the chance rate. So the model truly knows only about 6% of these facts; the rest of its correct picks are lucky guesses.

Does a family name mean an easier fact? We test whether facts written at the family level are simply easier. Qwen2.5-7B answers the same 1,800 facts closed book (Table 3).

Within each relation we build a 2×2 table (family-only vs. named, or family-only vs. silent; correct vs. wrong) and pool the three tables with the Mantel–Haenszel odds ratio (Appendix C). Family-only facts are much harder than named facts (OR = 0.27 [0.17, 0.43]) and as hard as silent facts (OR = 1.10 [0.80, 1.53]). An exact name in the text goes with 3.7 times higher odds that the model knows the fact; a family name goes with none.

C Statistical Procedures

This appendix defines every statistic in the paper. All tests are two-sided, and all code uses only the Python standard library, with random seed 0.

Page effect. For a question q and an outcome $o \in \{\text{correct}, \text{wrong}, \text{decline}\}$, let $y_q^{o,\text{page}} = 1$ if the model's answer has outcome o when it reads the page, and 0 otherwise. Let $y_q^{o,\text{closed}}$ be the same without the page. The page effect of q is

$$d_q^o = y_q^{o,\text{page}} - y_q^{o,\text{closed}} \in \{-1, 0, 1\}. \tag{3}$$

A page effect of +1 for declines means the page turned an answer into a decline; −1 means the page turned a decline into an answer. Our main measure (Eq. 1) is the mean page effect on family-only questions F minus that on silent questions S:

$$\Delta^o = \bar{d}_F^o - \bar{d}_S^o, \qquad \bar{d}_G^o = \frac{1}{|G|} \sum_{q \in G} d_q^o. \tag{4}$$

This is a difference in differences. The first difference (page minus closed book) removes how hard each fact is on its own. The second (family-only minus silent) asks what the family name adds beyond a page that says nothing.

Pooling models. For the pooled 7–8B result, we first average each question's page effect over Qwen2.5-7B, OLMo 2 7B and Granite 3.1 8B, and then compute Δ^o on these averages. Each of the 481 questions thus counts once, however many models answer it.

Confidence intervals. We draw 5,000 bootstrap samples. Each sample redraws the questions of each group with replacement, keeping the group sizes, and recomputes Δ^o. The 95% CI is the 2.5th to 97.5th percentile of these values. If the interval excludes zero, we call the effect reliable.

***p*-values.** We test $\Delta^o = 0$ with a Welch-type z-test, which allows the two groups to have different variances:

$$z = \frac{\bar{d}_F^o - \bar{d}_S^o}{\sqrt{s_F^2/|F| + s_S^2/|S|}}, \tag{5}$$

where s_G^2 is the sample variance of d_q^o in group G. The p-value is $2\,(1 - \Phi(|z|))$, with Φ the standard normal distribution function. A small p means a gap this large is unlikely if the family name had no effect. Each comparison covers three outcomes, so the Bonferroni-corrected p-value is three times the raw one.

Odds ratios. For the closed-book check in Appendix B, each relation r gives a 2×2 table: a_r and b_r are correct and wrong answers in the first group, c_r and e_r in the second, and n_r is the total. The Mantel–Haenszel odds ratio pools the relations:

$$\text{OR}_{\text{MH}} = \frac{\sum_r a_r e_r / n_r}{\sum_r b_r c_r / n_r}. \tag{6}$$

OR = 1 means the two groups are answered correctly equally often. OR = 0.27 means the odds of a correct answer in the first group are about a quarter of those in the second. Pooling within relations keeps the result from being driven by one relation that happens to be easier.

AUROC. To judge confidence filtering, we take the answers a model gives (not its declines) and ask how well the probability of the chosen option ranks correct answers above wrong ones. AUROC is the chance that a random correct answer has higher confidence than a random wrong answer. An AUROC of 1.0 means a threshold can separate them perfectly; 0.5 means confidence carries no signal.

McNemar's test. In recovery, two choosers answer the same 410 facts. Let b be the number of facts the first gets right and the second gets wrong, and c the reverse. McNemar's test uses only these disagreements, with a continuity correction:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}, \tag{7}$$

compared with a χ^2 distribution with one degree of freedom. It asks whether one chooser wins the disagreements more often than chance would allow.

D Abstention Questions and Full Results

This appendix shows how we build each question, how the model sees it, and how we read its answer. It then gives the full results behind Table 4 and Figure 2 of the main paper.

D.1 Building the Questions

Sampling. We sort PrimeKG facts into six cells: three page types (named, family-only, silent) times two relations (indication, contraindication). We shuffle each cell with seed 0 and take facts in order until the cell has 150 questions. We skip a fact when the drug has no text, when we cannot draw four wrong options, or when the 1,800-character page no longer shows the matched name or family. Only 106 family-only contraindications pass these checks, so we use all of them. This gives 856 questions.

Page type	Relation	n	All-neighbor (%)
Named	both	300	35.3
Family-only	contraindication	106	83.0
Family-only	indication	150	84.7
Silent	contraindication	150	45.3
Silent	indication	150	61.3

Table 4: Questions per group and the share whose four wrong options are all siblings or cousins of the answer in MONDO.

Question templates. Indication questions ask: *Which disease is X used to treat?* Contraindication questions ask: *In patients with which disease must X not be given?*

Wrong options. Each question has five diseases and the option *Not enough information to tell*, labeled A to F. We place the true disease and the decline option at random letters. We draw the four wrong options in this order:

1. *siblings*: other children of the answer's MONDO parent;
2. *cousins*: other grandchildren of the answer's grandparent;
3. *random* diseases from the same relation, only if siblings and cousins run out.

We never use an ancestor or descendant of the answer, or any other disease the drug has under the same relation in PrimeKG, since that option would also be correct. Because siblings and cousins share the answer's family, a page that names only the family cannot rule them out. Random diseases, however, can be ruled out by the family name alone, which makes a question easier. Table 4 shows that family-only questions more often have four neighborhood options than the other groups. This is why the main analysis keeps only questions whose four wrong options all come from the neighborhood (106 named, 215 family-only, 160 silent; 481 in all).

Windowed pages. A full drug article is too long for a prompt, so we cut a window around the evidence. The page starts with the first 800 characters of the drug's cleaned text. We then add a 280-character window on each side of every mention of the answer or its family. We merge windows that overlap, join them with ellipses, and stop at 1,800 characters. A silent page has no mentions, so it is just the opening of the article. Every model sees the same page for the same question.

Prompt. Each prompt is one user turn in the model's own chat template. The closed-book prompt is the same without the passage block:

```
Passage:
<passage>
Question: <question>
Options:
A. <option>
...
F. <option>
Answer with the letter of one option
only.
```

Reading the answer. We do not generate text. We read the probability of the next token for each letter. For letter ℓ, let $p(\ell)$ be the larger of the probabilities of "ℓ" and "ℓ" (with a leading space). We renormalize over the six letters:

$$\pi(\ell) = \frac{p(\ell)}{\sum_{m \in \{A,\ldots,F\}} p(m)}, \tag{8}$$

and the model's answer is the letter with the highest π. Its confidence is that π. If the answer is *Not enough information to tell*, the question counts as a decline. Precision is the share of correct answers among the questions the model does not decline.

D.2 Full Results

Without a page. Table 5 gives closed-book outcomes on the 481 matched questions. Family-only questions already draw more wrong answers and fewer declines than silent ones in every 7–8B model. For example, Qwen2.5-7B declines 32.1% of family-only questions and 59.4% of silent ones before it reads anything. If we compared pages directly, part of the gap would come from these facts, not from the page. The page effect removes this part.

Model	Group	Correct	Wrong	Declines
Qwen2.5 7B	named	60.4	17.9	21.7
	family-only	35.3	32.6	32.1
	silent	25.0	15.6	59.4
OLMo 2 7B	named	65.1	31.1	3.8
	family-only	36.7	56.7	6.5
	silent	37.5	51.2	11.2
Granite 3.1 8B	named	50.9	24.5	24.5
	family-only	33.0	33.0	34.0
	silent	26.2	26.2	47.5
Qwen2.5 0.5B	named	39.6	56.6	3.8
	family-only	30.2	69.8	0.0
	silent	28.7	68.8	2.5

Table 5: Closed-book outcomes (%) on the 481 neighborhood-matched questions (106 named, 215 family-only, and 160 silent) when no passage is provided.

With a page, all questions. Table 6 repeats Table 4 on all 856 questions, without the neighborhood filter. The pattern is the same: family-only pages draw far more wrong answers than silent pages, with lower precision. Compared directly with the page, family-only pages give more wrong answers than silent pages by 15.6, 20.3 and 16.5 points for Qwen2.5-7B, OLMo 2 and Granite ($p = 3\times10^{-6}, 5\times10^{-5}, 6\times10^{-5}$), and fewer declines by 28.4, 26.9 and 31.7 points ($p < 10^{-7}$ each).

Page effects per model. Table 7 breaks the page effect into its parts. For each outcome it shows the mean page effect on family-only questions, the same on silent questions, and their difference with its CI and p-value. The pooled row is the result in the main paper. OLMo 2 and Granite each show a large drop in declines; Qwen2.5-7B shows none, because

Model	Passage	Corr.	Wrong	Decl.	Prec.	Conf.
Qwen2.5 7B	named	64.7	3.3	32.0	95.1	0.99
	family-only	29.3	17.6	53.1	62.5	1.00
	silent	13.0	2.7	84.3	83.0	0.99
OLMo 2 7B	named	71.3	20.7	8.0	77.5	0.80
	family-only	39.1	42.6	18.4	47.8	0.84
	silent	32.0	28.0	40.0	53.3	0.76
Granite 3.1 8B	named	64.3	4.7	31.0	93.2	0.67
	family-only	33.2	25.0	41.8	57.0	0.93
	silent	15.7	11.0	73.3	58.8	0.81
Qwen2.5 0.5B	named	78.7	14.3	7.0	84.6	0.43
	family-only	37.5	52.3	10.2	41.7	0.55
	silent	28.7	50.0	21.3	36.4	0.39

Table 6: Open-book outcomes across all 856 PrimeKG questions (%; 300 named, 256 family-only, and 300 silent). Precision is the share of correct answers among non-abstained questions; Conf. is the median probability assigned to wrong answers.

it already declines family-only questions much less often closed book (Table 5).

By relation. Table 8 splits wrong answers with the page by relation. Family-only pages give more wrong answers than silent pages for both relations in every 7–8B model. For indications, Qwen2.5-7B is wrong on 26.0% of family-only questions and 2.2% of silent ones. Models are more careful on contraindications and decline more of them, but the gap stays. Qwen2.5-0.5B rarely declines at all (8–21% with a page, under 4% closed book), so its wrong-answer rates are high everywhere.

Family size. Table 9 splits the family-only questions into thirds by the size of the named family. The pooled drop in declines is about 17–18 points for families of up to 20 members and only 6.5 points, not significant, for larger families. A short list looks answerable to the model; a long one does not.

E Page Edits

The abstention test compares different facts: the family-only questions are not the same facts as the silent ones. The page edits remove this difference. They ask each question three times, with three versions of one page, and change nothing else.

Which questions. We start from named questions, whose page names the answer. In PrimeKG we keep the 130 whose answer has a MONDO parent, since the family edit needs a parent name. In GloBI we keep the 255 named questions whose answer has a genus.

Three versions.

- *Original*: the page as retrieved. It names the answer.
- *Family*: every mention of the answer, and of each of its alternative names, becomes the name of its immediate parent (the MONDO parent, or the genus). In GloBI we also replace the abbreviated form (*A. plantaginifolia*).
- *Removed*: every sentence that mentions the answer is deleted.

We replace whole words only. If the answer has several parents, we use the shortest name, and we never use a parent that is itself one of the options. We drop a question if the answer is still mentioned after the edit, or if nothing is left after removal. For example, a page that says "*. . . used to treat familial hyperlipidemia*" becomes "*. . . used to treat hyperlipidemia*" in the family version, and loses that sentence in the removed version. The question, the options, their letters and the model stay the same. Each prompt is cached under a hash of its exact text, so an edited page can never reuse the answer of the original.

What the comparison shows. The family and removed versions both lack the answer. The only difference is that the family version keeps the family name in place of the answer. If the family name had no effect, the two versions would give the same outcomes. Table 10 shows that they do not: in every model and both domains, the family version draws fewer declines than the removed version.

Pooled effects. Table 11 pools the three models, averaging each question over models first. Against the removed version, the family name cuts declines by 17.4 points in PrimeKG and 27.7 in GloBI, and raises both wrong and correct answers. Against the original, the family version loses many correct answers and gains wrong ones. So the family name lets the model guess among the family members without telling it which one is meant.

F Three Fixes

This appendix gives the per-model numbers behind Figure 4 of the main paper.

Confidence. A simple guard is to answer only when the model is confident. Table 12 shows, for each model and page type, how many questions the model answers, its precision, its mean confidence on right and on wrong answers, and the AUROC (Appendix C). On named pages, right answers carry clearly more confidence than wrong ones, and AUROC is 0.73–0.87. On family-only pages the two are close, and AUROC drops to 0.59–0.72. The wrong guesses are confident: Qwen2.5-14B puts a mean probability of 0.99 on its wrong family-only answers. In GloBI the drop is larger, from 0.82–0.86 on named pages to 0.52–0.68.

Scale. Within Qwen2.5, a larger model is more precise on named pages: 84.5% at 0.5B, 95.0% at 7B and 96.0% at 14B. On family-only pages, precision rises from 38.1% to 59.4% between 0.5B and 7B and then stops: the 14B model reaches 58.4%. The 14B model gives 14.7 points more wrong answers on family-only pages than on silent ones ($p = 6\times10^{-6}$). Like the 7B model, it already declines family-only questions less often closed book, so its page effect does not differ between the groups (+5.3 points, $p = 0.29$). A larger model reads named pages better but still guesses from a family name.

	Declines (Abstention)			Correct Answers			Wrong Answers		
Model	Fam.	Sil.	Diff. [95% CI], p	Fam.	Sil.	Diff. [95% CI], p	Fam.	Sil.	Diff. [95% CI], p
Qwen2.5 7B	+23.3	+24.4	−1.1 [−10.5, 8.4], .82	−8.8	−11.2	+2.4 [−5.7, 10.5], .57	−14.4	−13.1	−1.3 [−9.4, 7.1], .76
OLMo 2 7B	+11.6	+33.8	−22.1 [−31.8, −12.6], 6e-6	+0.5	−6.9	+7.3 [−2.6, 17.0], .15	−12.1	−26.9	+14.8 [3.6, 26.1], .009
Granite 3.1 8B	+9.3	+27.5	−18.2 [−28.5, −8.1], .0006	−2.8	−11.2	+8.5 [−0.9, 17.8], .064	−6.5	−16.2	+9.7 [−0.1, 19.9], .056
Qwen2.5 0.5B	+9.8	+18.1	−8.4 [−15.7, −1.1], .027	+4.2	0.0	+4.2 [−6.1, 14.4], .42	−14.0	−18.1	+4.2 [−6.4, 14.6], .44
Pooled 7–8B	+14.7	+28.5	−13.8 [−20.3, −7.5], 4e-5	−3.7	−9.8	+6.1 [−0.1, 12.4], .058	−11.0	−18.8	+7.7 [1.2, 14.0], .019

Table 7: Within-question page effects (in percentage points): the change in each outcome when adding a passage relative to closed-book answering on family-only (Fam., $n = 215$) and silent (Sil., $n = 160$) neighborhood-matched questions, along with the difference-in-differences estimate.

	Indication		Contraindication	
Model	Family-only	Silent	Family-only	Silent
Qwen2.5 7B	26.0	2.2	6.8	2.9
OLMo 2 7B	38.6	21.7	53.4	27.9
Granite 3.1 8B	37.8	16.3	10.2	1.5
Qwen2.5 0.5B	46.5	46.7	69.3	55.9

Table 8: Open-book wrong-answer rates (%) broken down by clinical relation on neighborhood-matched questions (indication: 127 family-only, 92 silent; contraindication: 88 family-only, 68 silent).

Family size	n	Median	Decline effect [95% CI]	p
≤ 10	71	7	−17.3 [−26.2, −8.2]	.001
11–20	69	16	−18.4 [−26.6, −10.3]	<.001
> 20	68	44	−6.5 [−15.7, +3.0]	.174

Table 9: Decline effect (Δ for declines, points) by the number of members in the named family, pooled over the three 7–8B models.

Instruction. We add one sentence to the prompt with the page:

> *If the passage names only a broad category of diseases and does not say which specific disease is meant, answer with the letter of "Not enough information to tell.*

Table 13 shows the result. The models react very differently. OLMo 2 declines far more on every page type, including named pages (from 7.5% to 27.4%). Granite barely reacts. Qwen2.5-7B declines somewhat more. Pooled over the three models, the instruction raises declines by 15.2 points on family-only pages, but also by 14.0 on silent pages and 9.1 on named pages, where it costs 6.3 points of correct answers. Precision on family-only pages stays low (47.9–64.9%). The instruction makes the models more cautious everywhere, but it does not teach them to tell a family name from an answer.

G Second Domain: GloBI

We repeat the abstention test and the page edits on species interactions. This tests whether CLADE depends on medicine, on MONDO, or on PrimeKG. It does not: the same pattern appears with a different graph, a different hierarchy and different text.

Model	Version	Decl.	Corr.	Wrong	Prec.
PrimeKG (130 questions)					
Qwen2.5-7B	original	27.7	66.9	5.4	92.6
	family	43.8	46.2	10.0	82.2
	removed	70.0	26.9	3.1	89.7
OLMo 2 7B	original	6.2	74.6	19.2	79.5
	family	7.7	58.5	33.8	63.3
	removed	18.5	52.3	29.2	64.2
Granite 3.1 8B	original	23.8	69.2	6.9	90.9
	family	40.8	43.1	16.2	72.7
	removed	56.2	31.5	12.3	71.9
GloBI (255 questions)					
Qwen2.5-7B	original	38.4	58.8	2.7	95.5
	family	71.0	21.6	7.5	74.3
	removed	92.2	5.1	2.7	65.0
OLMo 2 7B	original	17.3	71.4	11.4	86.3
	family	25.9	44.3	29.8	59.8
	removed	56.5	20.8	22.7	47.7
Granite 3.1 8B	original	32.2	61.2	6.7	90.2
	family	50.2	32.9	16.9	66.1
	removed	81.6	9.8	8.6	53.2

Table 10: Outcomes (%) with each version of the page. In every model and domain, the family version draws fewer declines than the removed version.

G.1 Setup

Facts and pages. Each fact links a consumer species to a partner species, such as a butterfly and the plant its larvae eat. The page is the consumer's Wikipedia article, cleaned as in Appendix B. A page is *named* if it gives the partner's Latin name, its abbreviated form (*A. plantaginifolia*) or a common name from Wikipedia redirects. It is *family-only* if it gives only the partner's genus or family, and *silent* if it gives neither. We measure 23 694 edges over 5425 articles. Pages name the partner for only 3.9% of edges and name only its genus or family for 7.1% (Table 3). Family-only pages are thus about twice as common as named ones.

Questions. We use one template per relation:

	Outcome	Family − removed	Family − original
PrimeKG	Declines	−17.4 [−23.6, −11.5]	+11.5 [+7.4, +15.9]
	Wrong	+5.1 [+0.5, +10.3]	+9.5 [+5.9, +13.6]
	Correct	+12.3 [+6.9, +17.9]	−21.0 [−26.9, −15.6]
GloBI	Declines	−27.7 [−32.2, −23.5]	+19.7 [+15.4, +23.9]
	Wrong	+6.7 [+3.4, +10.2]	+11.1 [+8.2, +14.1]
	Correct	+21.0 [+16.9, +25.4]	−30.8 [−36.2, −25.5]

Table 11: Pooled edit effects over the three 7–8B models (points, 95% bootstrap CI).

Model	Page	Ans.	Prec.	Conf. R	Conf. W	AUROC
PrimeKG						
Qwen2.5-0.5B	named	97	84.5	.83	.49	.85
	family	194	38.1	.71	.60	.64
Qwen2.5-7B	named	80	95.0	.98	1.00	.73
	family	96	59.4	.98	.90	.72
OLMo 2 7B	named	98	83.7	.97	.75	.86
	family	176	45.5	.89	.77	.71
Granite 3.1 8B	named	85	90.6	.95	.71	.87
	family	122	53.3	.90	.82	.66
Qwen2.5-14B	named	75	96.0	.99	.92	.77
	family	89	58.4	.98	.99	.59
GloBI						
Qwen2.5-7B	named	167	94.6	.98	.90	.82
	family	18	27.8	.94	.89	.55
OLMo 2 7B	named	231	84.4	.94	.69	.86
	family	138	26.8	.76	.62	.68
Granite 3.1 8B	named	188	89.4	.94	.75	.86
	family	46	21.7	.75	.74	.52

Table 12: Confidence of answers with the page. Ans. is the number of questions answered (not declined); Conf. R and Conf. W are the mean confidence of right and wrong answers.

Relation	Question
eats	Which species does X feed on?
preys on	Which species does X prey on?
parasite of	Which species is X a parasite of?
pollinates	Which plant does X pollinate?

We take 75 questions per page type and relation, for 900 in all. Wrong options are other species of the partner's genus first, then of its family, and never another recorded partner of the consumer. For binomial answers, we fill any remaining slots with species from other genera, never with a bare genus name, so every option is a species. Most questions have four neighborhood options (833 of 900; Table 14), and the main analysis keeps only these.

G.2 Results per Model

Outcomes. Table 15 gives outcomes for each model, closed book and with the page. Closed book, all three models decline most species questions; they know few of these facts. A named page lets them answer, and their answers are precise (84–95%). A family-only page also lets them answer, but most of these answers are wrong: precision is 22–28%, close to the 20% chance rate of a five-way guess. A silent page makes them decline almost everything.

		Declines		Precision	
Model	Page	base	instr.	base	instr.
Qwen2.5-7B	named	24.5	29.2	95.0	94.7
	family	55.4	64.2	59.4	64.9
	silent	83.8	88.1	84.6	89.5
OLMo 2 7B	named	7.5	27.4	83.7	89.6
	family	18.1	56.3	45.5	47.9
	silent	45.0	85.6	55.7	60.9
Granite 3.1 8B	named	19.8	22.6	90.6	91.5
	family	43.3	41.9	53.3	53.6
	silent	75.0	71.9	60.0	55.6

Table 13: Declines and precision (%) with the page, without (base) and with (instr.) the instruction to decline on a broad category.

Page type	eats	parasite of	pollinates	preys on
Named	93.3	94.7	94.7	89.3
Family-only	100.0	98.7	100.0	93.3
Silent	90.7	85.3	92.0	78.7

Table 14: Share of GloBI questions (%, 75 per cell) whose four wrong options are all from the partner's genus or family.

Page effects. Table 16 gives the page effect for each model. The effect is largest in OLMo 2, where a silent page raises declines by 18.5 points but a family-only page lowers them by 13.3, a difference of 31.7 points. Qwen2.5-7B declines almost every question on either page, so its effect is small but still significant. Granite's decline effect is not significant alone, but its wrong answers rise reliably. Pooled over the three models, $\Delta = -14.3$ points for declines and +11.6 for wrong answers, with only +2.7 for correct answers: in GloBI, the answers that the family name frees are mostly wrong.

Page edits and confidence. The per-model edit results are in Table 10 and the confidence results in Table 12. In every model, the genus name in place of the answer draws fewer declines than deleting the sentence, and AUROC on family-only pages falls to 0.52–0.68.

Why GloBI is harder. Sibling species are harder to tell apart than sibling diseases. The models know few species facts closed book (Table 15), so when a page names only the genus, they guess inside the genus close to chance. This is why precision on family-only pages is near chance in GloBI (22–28%) but closer to half in PrimeKG (45–59%).

H Ontology Recovery

Recovery asks whether the ontology can turn a family name back into the exact answer. It has three steps: build a list of candidates, choose one, and compare with baselines that do not use the ontology.

Facts. We use all 410 PrimeKG indications and contraindications whose combined Wikipedia and PubMed text names only a MONDO family that expands to at least two diseases.

		Correct		Wrong		Declines	
Model	Page	cl.	page	cl.	page	cl.	page
Qwen2.5-7B	named	7.9	56.6	4.3	3.2	87.8	40.1
	family	1.7	1.7	2.7	4.4	95.6	93.9
	silent	3.5	0.4	1.2	1.2	95.4	98.5
OLMo 2 7B	named	20.8	69.9	24.7	12.9	54.5	17.2
	family	10.9	12.6	22.8	34.4	66.3	53.1
	silent	8.5	3.5	22.3	8.8	69.2	87.7
Granite 3.1 8B	named	12.2	60.2	16.8	7.2	71.0	32.6
	family	9.2	3.4	14.6	12.2	76.2	84.4
	silent	5.8	1.5	12.3	1.9	81.9	96.5

Table 15: GloBI outcomes (%) closed book (cl.) and with the page, on questions whose four wrong options all come from the neighborhood (279 named, 294 family-only, 260 silent).

Model	Outcome	Δ	[95% CI]	*p*
Qwen2.5-7B	declines	−4.8	[−8.9, −0.8]	.022
	wrong	+1.7	[−1.1, +4.8]	.26
OLMo 2 7B	declines	−31.7	[−40.6, −22.7]	2e-11
	wrong	+25.0	[+16.5, +33.4]	1e-8
Granite 3.1 8B	declines	−6.5	[−14.0, +0.8]	.097
	wrong	+8.0	[+1.5, +14.6]	.019
Pooled	declines	−14.3	[−18.7, −9.6]	1e-9
	wrong	+11.6	[+7.5, +15.4]	3e-8

Table 16: GloBI page effects (points), family-only minus silent (Eq. 1).

A drug often has several true diseases in the same family (2.47 per list on average). We therefore count a pick as correct if it matches any disease the drug has under that relation. Table 19 also gives the strict score, which accepts only the one sampled answer.

H.1 Building the List

Given family. This list assumes we already know which family on the page belongs to the question. We start at the true disease and climb up to two levels in MONDO. Among these ancestors, we keep the ones the drug's text names. For each, we collect its descendants up to two levels down. We keep the ancestor with the smallest list, sort the list, and cap it at 64 diseases. This is an oracle: it uses the true disease to pick the family. It tells us how good the list can be at best.

Page built. This list uses only the page, as a real system would. We take all 3,182 MONDO families with two or more children as candidate family names. We index every run of up to eight words on the drug's cleaned Wikipedia page, adding singular forms of words that end in *s*. Every family whose name appears in this index is a match. We drop any matched family whose own two-level list exceeds 64 diseases, and join the lists of the rest. If the joined list still exceeds 64, we mark the page as *too broad*. In short:

```
fams = [f for f in families if name(f)
in page]
fams = [f for f in fams if |list(f)| <=
64]
cands = union(list(f) for f in fams)
if |cands| > 64: too broad
```

	Wikipedia	+ PubMed
Matched families per page (median)	5	17
Usable lists (of 410)	166	19
Lists over the 64 cap	209	391
Usable list holds sampled answer	19.8%	3.7%
Usable list holds any true answer	24.6%	4.1%
Median usable list size	32	59
End-to-end recovery (evidence pick)	10.0%	1.5%

Table 17: Page-built lists from Wikipedia alone and from Wikipedia plus PubMed, over all 410 family-only facts. The remaining unusable lists match no family with two or more children.

Why page-built lists fail. A drug article names many disease families that are not the answer's: other uses, side effects, warnings and conditions that often occur together. The median Wikipedia page matches five families; with PubMed it matches 17. Their union is usually too large (Table 17). The family that holds the answer is on the page, but the matcher cannot tell it apart from the others.

H.2 Choosing One Candidate

Model scores. Qwen2.5-7B scores each candidate by its mean log-probability as the completion of a relation stem, such as "*X is a drug used to treat the disease called*". In the *model alone* condition the prompt has only the stem; in the *model with page* condition the page comes first. The highest score is the pick.

Evidence scores. We use two outside sources. The *Wikipedia reverse lookup* checks whether the candidate disease's own article names the drug (a *mention*), and whether a relation keyword appears within 250 characters of that mention (a *cue*). Indication keywords are *treat*, *therap*, *indicated* and *first-line*; contraindication keywords are *contraindicat*, *avoid*, *must not*, *caution*, *worsen* and *adverse*. The *PubMed co-mention* count is the number of records that name both the drug and the candidate with a relation qualifier:

```
"<drug>"[tiab] AND "<cand>"[tiab] AND
<q>
```

where the indication qualifier `<q>` is `(treat*[tiab] OR therap*[tiab] OR efficac*[tiab])` and the contraindication qualifier is `(contraindicat*[tiab] OR adverse[tiab] OR toxicit*[tiab] OR "side effect"[tiab] OR "side effects"[tiab] OR induced[tiab])`. We strip parenthetical parts such as "(disease)" from MONDO labels before we query. The combined evidence score of candidate c is

$$E(c) = 2 \cdot \mathbf{1}[\text{cue}] + \mathbf{1}[\text{mention}] + \log\big(1 + \text{co-mentions}\big). \quad (9)$$

A signal makes a pick only when one candidate has the strictly highest score; otherwise it records a tie or no signal.

Signal	Picks	Ties	None	Acc./pick	Acc./all
Wikipedia lookup	82	80	248	50.0	10.0
PubMed co-mention	324	12	74	45.4	35.9
Combined evidence	327	14	69	44.6	35.6
Popularity prior	409	0	1	46.9	46.8
Normalized PubMed	336	0	74	31.0	25.4

Table 18: Signals inside the given-family list (410 facts, 8448 candidates). Accuracy (%) counts a pick as correct if it matches any true disease of the drug.

Condition	Sampled	Any true
Model, given-family list	18.3	35.9
Model + page, given-family list	17.8	32.4
Model, random list	30.5	34.9
Model + page, random list	29.0	32.7

Table 19: Qwen2.5-7B recovery accuracy (%) on the 410 family-only facts, scoring only the sampled answer (Sampled) or any true answer of the drug (Any true).

The *evidence, then model* chooser uses E when it makes a pick and falls back to the model otherwise.

Popularity prior. The popularity prior ignores the drug. It counts all PubMed records that name the candidate (`"<cand>"[tiab]`) and picks the candidate with the most. The *normalized* co-mention signal divides each co-mention count by this total, which removes the popularity part.

What each signal does. Table 18 shows how often each signal makes a pick and how accurate those picks are. The Wikipedia lookup is accurate when it fires (50.0%), but it fires for only 82 facts, because a disease article rarely names a specific drug. PubMed co-mentions fire often, but they are only as accurate as the popularity prior. Once we divide out popularity, accuracy falls from 35.9% to 25.4%. Most of the signal in co-mentions is therefore how much a disease is studied, not its link to the drug.

H.3 Baselines Without the Ontology

To see what the ontology adds, we replace the family list with a list of the same length that holds the sampled answer and random diseases from the same relation (never another true partner of the drug). The model scores this *random list* with or without the page. Table 19 compares the conditions. Under the strict score, the random list looks easier, because its wrong candidates are unrelated diseases that the model can rule out. Under the any-true-answer score, the family list catches the drug's other true diseases, and the model does about as well on it as on the random list. The family list is valuable because it is short and complete (98.5% coverage), not because it makes the model's choice easier.

H.4 Examples

Table 20 shows evidence picks within the given-family list. When the pick misses the PrimeKG answer, it usually takes a better-studied member of the same family (*ovarian carcinoma* for carboplatin) or a true use that PrimeKG does not list (*Legionnaires' disease* for moxifloxacin, which is used against this pneumonia). Strict scoring against the graph counts both as wrong, so it understates how often the pick is clinically sound.

Drug	PrimeKG Target	Evidence Pick	Match?
Lovastatin	familial hyperlipidemia	familial hyperlipidemia	yes
Tetrabenazine	juvenile Huntington disease	juvenile Huntington disease	yes
Moxifloxacin	Klebsiella pneumonia	Legionnaires' disease	no
Carboplatin	malignant germ cell tumor of ovary	ovarian carcinoma	no

Table 20: Qualitative examples of candidate diseases chosen by combined literature evidence within the given-family list on indication facts.

I Worked Examples

This appendix follows single questions through the test. Each example shows the page, the question and what each model answers, closed book and with the page. Page excerpts come from English Wikipedia (CC BY-SA 4.0); "..." marks text we leave out.

I.1 PrimeKG

Moxifloxacin: a confident wrong guess.

Fact: moxifloxacin treats *Klebsiella pneumonia*.
Page type: family-only. The page says the drug treats *pneumonia*, but not which kind.
Question: Which disease is moxifloxacin used to treat?
Options include: Klebsiella pneumonia, Legionnaires' disease, pneumonia caused by gram negative bacteria, and other bacterial pneumonias; Not enough information to tell.

Every model answers, and every model is wrong:

Model	Closed book	With the page (conf.)
Qwen2.5-0.5B	gram-negative pn.	gram-negative pn. (.35)
Qwen2.5-7B	Legionnaires'	Legionnaires' (.88)
OLMo 2 7B	gram-negative pn.	gram-negative pn. (.86)
Granite 3.1 8B	gram-negative pn.	Legionnaires' (.77)
Qwen2.5-14B	Legionnaires'	gram-negative pn. (1.00)

The page tells the models that the answer is a pneumonia, and they pick a pneumonia with high confidence. None of them declines, even though the page cannot say which pneumonia is meant. A confidence threshold would keep most of these answers. Note also that *Legionnaires' disease* is a real use of moxifloxacin that PrimeKG does not list: the graph counts a sound answer as wrong (Appendix H).

Lovastatin: the page makes one model worse.

Fact: lovastatin treats *hyperlipidemia, familial combined, LPL related*.
Page type: family-only. The page says *hyperlipidemia*.
Family in MONDO: hyperlipidemia → familial hyperlipidemia → {the answer, hyperlipidemia due to hepatic triglyceride lipase deficiency, familial lipoprotein lipase deficiency, . . . }.

Model	Closed book	With the page
Qwen2.5-0.5B	wrong sibling	wrong sibling
Qwen2.5-7B	declines	**correct** (1.00)
OLMo 2 7B	**correct** (.95)	wrong sibling (.54)
Granite 3.1 8B	**correct** (.98)	**correct** (.86)
Qwen2.5-14B	wrong sibling	wrong sibling (1.00)

Here the family name helps Qwen2.5-7B and hurts OLMo 2. OLMo 2 knows the answer closed book, but with the page it switches to *hyperlipidemia due to hepatic triglyceride lipase deficiency*, a sibling. The page points to the right family, and the model then picks the wrong member. This is CLADE in its plainest form.

I.2 GloBI

Named page: the answer is on the page.

Page (*Congriscus megastomus*): "*Congriscus megastomus is an eel in the family Congridae (conger/garden eels). . . . It is a marine, temperate-water dwelling eel which is known from Japan and the Kyushu–Palau Ridge . . .* " The page later names the prey.
Question: Which species does Congriscus megastomus feed on?
Options: Ventrifossa garmani, Ventrifossa mucocephalus, Coryphaenoides zaniophorus, *Hymenocephalus lethonemus* (true), Gadomus colletti, Not enough information to tell.

All three models decline closed book. With the page, Qwen2.5-7B and Granite pick the true species (confidence 1.00 and .997); OLMo 2 declines. When the page names the answer, the models use it well.

Family-only page: the page turns declines into guesses.

Page (*Vanessa virginiensis*, American painted lady): "*The larvae feed on various Asteraceae, such as the cudweeds (genus Gnaphalium), the pussytoes (**Antennaria**), and the everlastings (Anaphalis) . . .* "
Question: Which species does Vanessa virginiensis feed on?
Options: Antennaria pulcherrima, *Antennaria plantaginifolia* (true), Antennaria dimorpha, Antennaria lanata, Antennaria microphylla, Not enough information to tell.

Every option is an *Antennaria* species, so the genus name on the page rules out none of them. Closed book, all three models decline. With the page, Qwen2.5-7B still declines, but OLMo 2 picks *A. microphylla* (.52) and Granite picks *A. pulcherrima* (.63). Both are wrong. The page did not tell them the answer; it only made the question look answerable.

Silent page: the models abstain.

Page (*Cottunculus microps*, polar sculpin): "*. . . a species of fathead sculpin, a deepwater fish found in the North Atlantic and Arctic Oceans. . . . Like other members of its family, the polar sculpin has a large head, a short tapering body and a small tail. . . .* "
Question: Which species does Cottunculus microps feed on?
True answer: *Lumpenus lampretaeformis*.

The page describes the fish but says nothing about its food. All three models decline, closed book and with the page. This is the behavior we want on a family-only page as well, unless the system hands the question to the taxonomy.

A named page the models miss. The page for *Orgyia vetusta* (western tussock moth) names its food plant only by the common name *toyon*, which redirects to *Photinia arbutifolia*. Our matcher counts this page as named. All three models decline anyway: they do not link the common name to the Latin name in the options. Common names are a second gap between how text names a fact and how a graph stores it.

J Software and Compute

Models. We use the instruction-tuned checkpoints from Hugging Face: `Qwen2.5-0.5B-Instruct`, `Qwen2.5-7B-Instruct` and `Qwen2.5-14B-Instruct` (Qwen Team 2024), `OLMo-2-1124-7B-Instruct` (Team OLMo et al. 2024), and `granite-3.1-8b-instruct` (IBM Granite Team 2024). We do not fine-tune any model.

Running the models. All models run in 32-bit floating point with Hugging Face Transformers and PyTorch, on CPU nodes of a university cluster. Each model answers all 1,712 abstention prompts (856 questions, closed book and with the page) in one 12-thread job. We score prompts in batches with left padding. We rebuild the position indices from the attention mask, so a batched prompt gets exactly the same probabilities as the same prompt alone. Every answer is stored under a hash of its exact prompt, so an edited page never reuses an old answer.

Recovery. In recovery, all candidates for one fact share the same prompt prefix. We encode this prefix once and reuse its key–value cache for every candidate. Before each run, we check a sample of these cached scores against full forward passes.

Randomness and statistics. All sampling uses seed 0, and the bootstrap draws are sorted by question identifier before resampling, so every interval in the paper is the same on every run. The statistics use only the Python standard library.

K Choosing the Family and Free-Text Answers

This appendix covers two follow-up tests. The first tests a simple fix for step 2 of the hand-off: which family on the page holds the answer. The second asks the abstention questions without options, to check that the option list does not drive the main result.

K.1 Choosing the Family on the Page

Setup. We use the same 410 family-only facts as in Appendix H. For each fact, we take the families that the page-built list finds on the drug's Wikipedia page and keep those whose own list has at most 64 diseases. We order them by

Rule	Sampled answer	Any true answer
Qwen2.5-7B picks the family	**30.0**	**44.9**
First mentioned	25.1	37.3
Random	13.9	22.0
Smallest family	7.8	13.2
Page-built list (all families)	19.8	24.6
Any shown family (ceiling)	50.0	64.6

Table 21: Share of the 410 family-only facts (%) whose chosen list holds the sampled answer, or any true answer of the drug. At most ten families are shown per page.

where the page first mentions them and show at most ten, labeled A to J. Qwen2.5-7B reads a window of the page around these mentions and the question, and picks one family:

```
Passage:
<page window>
Question: Which disease is X used to
treat?
The answer is one specific disease.
Which of these disease categories from
the passage contains it?
Options:
A. <family>
...
Answer with the letter of one option
only.
```

We read the answer from the letter probabilities, as in Appendix D. We then expand only the chosen family, two levels down, and score its members with the same choosers as before.

Rules that read no meaning. To see what the model adds, we compare it with three rules that ignore the question: the family the page mentions first, the family with the smallest list, and a random family. We also report a ceiling: the share of facts where at least one of the shown families holds the answer.

Results. Table 21 shows the result. The model's choice holds the answer for 30.0% of all 410 facts, against 19.8% for the page-built list, which joins every family on the page. It also beats each rule. The first-mentioned family is a strong rule (25.1%); the smallest family is a poor one (7.8%), worse than a random pick. The ceiling is 50.0%: for the other half of the facts, the family that holds the answer is not among the usable families shown. The model therefore closes 30.0 of the 50.0 points that any chooser over these families could reach.

End to end. The chosen lists are short (median 12 diseases), and 375 of the 410 facts get one. Scored with the same choosers, they recover the answer for 19.3% of facts with the combined evidence and 26.8% with the popularity prior. The page-built lists reach 10.0% and 9.0%. With no oracle at any step, a single prompted choice of the family nearly triples end-to-end recovery.

Model	Page	Corr.	Abst.	Family	Sibl.	Other
Qwen2.5-7B	named	20.0	47.5	0.0	0.0	32.5
	family	0.0	42.5	17.5	0.0	40.0
	silent	0.0	52.5	0.0	0.0	47.5
OLMo 2 7B	named	25.0	37.5	0.0	0.0	37.5
	family	0.0	32.5	20.0	2.5	45.0
	silent	0.0	52.5	0.0	0.0	47.5
Granite 3.1 8B	named	22.5	37.5	0.0	0.0	40.0
	family	0.0	30.0	45.0	2.5	22.5
	silent	0.0	45.0	0.0	0.0	55.0

Table 22: Free-text outcomes with the page (%, 40 questions per row).

K.2 Free-Text Answers

Setup. We sample 40 questions per page type (named, family-only, silent) from the 481 questions whose four wrong options all come from the answer's neighborhood, with seed 0. We ask each one closed book and with the page, with no options:

```
Passage:
<page>
Question: Which disease is X used to
treat?
Answer with the name of one disease
only. If the passage does not tell
you which one, answer "Not enough
information to tell".
```

Each model writes up to 32 tokens with greedy decoding.

Scoring. We check each answer in this order and assign the first outcome that applies:

1. *correct*: it names the answer or one of its alternative names;
2. *abstain*: it says there is not enough information, or that the passage does not say;
3. *sibling*: it names one of the four wrong options of the multiple-choice question;
4. *family*: it names one of the answer's family names, such as *acute myeloid leukemia* for *acute myeloid leukemia with t(8;21)*;
5. *other*: it names any other disease.

The scoring is lenient: an answer that lists several diseases counts as correct if one of them is the answer. This affects only Granite on named pages, where 7 of its 9 correct answers are lists. It does not affect family-only or silent pages, where no model is ever correct.

Results. Table 22 gives the outcomes with the page. Three patterns stand out.

- *No exact answers from family-only pages.* No model ever names the exact answer from a family-only or silent page. When the page names the answer, the models name it 20.0–25.0% of the time.
- *Answering with the family.* On family-only pages, 17.5–45.0% of answers name the family itself, such as "acute myeloid leukemia (AML)". Closed book, this happens

10.0–15.0% of the time. It never happens on named or silent pages. Without options, CLADE takes this form: the model does not pick a sibling, but gives the coarse category as if it were the answer.

- *Fewer abstentions.* Compared with silent pages, family-only pages lower abstention in all three models. The page effects (Eq. 1) are −5.0 [−25.0, +15.0] for Qwen2.5-7B, −17.5 [−45.0, +7.5] for OLMo 2 and −10.0 [−32.5, +10.0] for Granite. All three point the same way as in the main test, but with 40 questions per group no single interval excludes zero.

Free text also makes the models more cautious overall. They abstain on 37.5–47.5% of named pages, where the multiple-choice test shows 7.5–24.5%. Writing the answer is harder than picking a letter, so we compare page types within the free-text format, not across formats.

References

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